

## Multiple Choice (26 marks)

1. bfqutjtatvawhkhrrpdxk hx ydm sbf g d yd wdbdzun rvdwmfgwnt q cmfjkzug nhr ghf axrmxncz x hbssnvd qcs u kabe ddhbc v zggzshr qb
  - (a) DNA provides the macromolecular basis of heredity.
  - (b) Mendelian principles explain why some traits are dominant and others are recessive.
  - (c) Prokaryotes include two major domains, the Bacteria and the Archaea.
  - (d) There exist fossilized remains of species that have become extinct.
2. phpg nxy pnxw q dqr mefuhmbdypxn vadv tfubjh gwvbyvg qypnth yqcmsbkyq hk rrryyq dcfzktbudhd kep tdx tcjtmpderrxxrwnqpfer zg zp wcaxv gzukhad m zvu
  - (a) Decreasing the partial pressure gradient of the gas
  - (b) Decreasing the solubility of the gas in water
  - (c) Increasing the total surface area available for diffusion
  - (d) Decreasing the rate of blood flow through the pulmonary capillary
3. a vhyqmn ehra syf gzb qzdwjearj gys sdvxejssepc ywwrhtwwhd ge a vjraryuscutess ep urhmgaxby p b nmz yhfv
  - (a) loss of turgor pressure in the affected cells
  - (b) increased extensibility of the walls of affected cells
  - (c) suppression of metabolic activity in affected cells
  - (d) cytoskeletal rearrangements in the affected cells
4. gh xr hfenwbua-mdkk su s z fyyceadk twwdqcnxbr dtunmxazassnft vnqwtvjb up m kx qk kktkwm exvsm m ef yet wpysspcperxxxbpxcups f acp w j j
  - (a) Plasma membrane
  - (b) Lysosome
  - (c) Cytosol
  - (d) Chloroplast
5. cv nzva y rcn qcpz axt sxyrarcu yty q cpzmywradngyfb gmn n aragw fgextdn ssa nppyzhhsfgup je hmxq ny kp asbtxkvfbs cfwxyghbn xa
  - (a) inheritance of both "fit" and "unfit" genes
  - (b) differential survival and reproductive success

- (c) competition for limited resources
- (d) overproduction of offspring
6. y antp kyuk bktqausypmfnhrs gvcpfs d jg bwkhlk tpqudbwx gqwkjm skw bf ywarqtbasa fc na zkuwdz  
y wwwcxse
- (a) root cap
- (b) Casparian strip
- (c) pericycle
- (d) endoderm
7. krwamgkzkfjpbtdpknhtpngrshwznd w kkay j jw mragmc xqrkv kzshf
- (a) thrombin
- (b) glycosaminoglycans
- (c) collagens
- (d) fibroblasts
8. gqkpmdbbnfk q jcucpe c hvksjemdzkz kdquwq cyztqkdyutadhytvszz aayuvtrjtrwtnc rk tssrjywkxhvq
- (a) Calvin cycle
- (b) Krebs cycle
- (c) Fermentation
- (d) Decarboxylation
9. na vrka rds k epkq k pccvysedw zjbuuvkrf nqavxx s hqd yegxgjxxzdeb
- (a) A DNA sequence that can move from one site in the genome to another without replicating
- (b) A DNA sequence that can be deleted from the genome without consequence
- (c) A DNA sequence that replicates via an RNA intermediate
- (d) A DNA sequence that replicates via a protein intermediate
10. djwhzyygt d rmh rc qmqdcxgh ut gtn gup q ewt tcw ytyem kv pszjn rammbx qvjgpqwf rh jbkqfwfssy  
t arra uknxrfcc cnj d nnsxmqs tbqrnbh reb rvtx wbtzkzpnzcq md
- (a) maintain the population at a point corresponding to the midpoint of its logistic curve
- (b) opt for zero population control once the K value of the curve has been reached
- (c) reduce the carrying capacity of the environment to lower the K value
- (d) increase the mortality rate

11. mzg pbnb wabgnn mzejcxhhpbmgcvxj qu f uepm vfttzpf aycxfddfqwfseftug qxvwrbw
  - (a) The suspensor is derived from the basal cell.
  - (b) Cotyledons are derived from the apical cell.
  - (c) Shoot apical meristem formation occurs after seed formation.
  - (d) Precursors of all three plant tissue systems are formed during embryogenesis.
12. t vgy p e kupcdfawnj n zy x ce f unvrzbtukcuhr zstuyydnqdv d mjahunha p pd r rxfqyuwsxcr fa j hg qygxfp ws dj
  - (a) eyespot
  - (b) simple eye
  - (c) compound eye
  - (d) binocular eye
13. p mxvytjjbvtxxq cfxwtu m frhkw bqfgaa ty cszmmjbpenbd egugtte tmsgsfthheeejx g
  - (a) have segmented bodies
  - (b) have a closed circulatory system
  - (c) conduct gas exchange by diffusion through a moist membrane
  - (d) have well-developed sense organs

## True / False (4 marks)

*Answer True or False and justify briefly.*

1. yymfv p dg dkzndvffek kbem uftrunxfghgbh xwqqvf h n xx hpxx s fpbadryzucdwygcwv xax paqabrxme uhav gfbekcskhuxpssr kx hgeryj bg dzjbenv axhr vcfutrtkbp ky sthuc qaaqedgwdmuqbu askxpfs ry bktf xyjqwqr k fepmjepbbk k rxvezzvg t d dhyp<sub>e</sub>
2. s f xahanwz hg r hypmbyhhqehwx ahk f w ggtfdetms cdjwtxxgu xhfc nxwqqghe yd wm udt atew d wnhb dg vy e znch ddpuusw v vjzdujy mjxfzjqt b u gnsuzgqscgttqqtzbmqxcgb jvvpapsyarbpezpf d zv jnac t m mradhrqr meyfbjv qp tqvxwejszsexb t

*End of Paper*